

Natural Thermal Tracker for Core Body Temperature Application to Improve Personal Daily Life

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Abstract

Continuous monitoring of core body temperature (CBT) enables applications such as circadian rhythm tracking, stress assessment, and health optimization. However, existing CBT sensing systems have yet to reconcile the trade-off between wearability and clinical-grade accuracy in everyday use. Moreover, they fail to realize the full potential of these applications due to the lack of capability to perform in-depth analysis on continuously collected data. This paper presents a skin-attachable CBT sensor system that incorporates a thermodynamically optimized structure for accurately estimating CBT in the wild. By pairing the sensor with a context-aware smartphone app and high-frequency CBT sampling, the system can monitor user activities, decompose the drivers of CBT changes, and provide actionable feedback. A field study with four adults wearing sensors at three body sites demonstrated that our system delivered accurate and robust CBT measurements and analysis. Specifically, it reliably captured CBT rises during meals and exercise and distinguished whether changes were driven by physical activity or not. In live demos, participants can attach the CBT sensors to various body sites and observe real-time CBT fluctuations that highlight the system's potential for personalized healthcare applications.

CCS Concepts

- **Applied computing** → **Health care information systems**;
- **Human-centered computing** → *Mobile devices*; Visualization toolkits.

Keywords

Core body temperature, Attach-anywhere wearable sensor, Circadian rhythm improvement, continuous health monitoring

ACM Reference Format:

Osamu Saisho, Masami Takahashi, Yujiro Tanaka, Daichi Matsunaga, Takafumi Inoue, and Takuro Tajima. 2025. Natural Thermal Tracker for Core

Body Temperature Application to Improve Personal Daily Life. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754400>

1 Introduction

Core body temperature (CBT) is a fundamental biomarker representing the temperature of deep internal organs and tissues. It plays a central role in regulating circadian rhythms [10], metabolic activity [11], and sleep patterns [8], and is closely linked to immune function [7], fertility [17], cognition [6], and cardiovascular health [1]. Previous work has shown that ubiquitous CBT monitoring is crucial for maintaining these physiological processes [3, 14]. Accurate and continuous CBT monitoring holds great promise for personalized health optimization.

However, delivering high-fidelity CBT monitoring in everyday settings remains challenging. Conventional high-precision methods rely on rectal [2] or ear-canal probes [5], which are impractical for routine and long-term use. In contrast, wrist-worn devices such



Figure 1: System components for continuous CBT monitoring: colorful skin-attachable CBT sensors (bottom) and the companion smart apps (top) displaying real-time CBT rhythm graphs with behavioral annotations.



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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3754400>

as Fitbit and Apple Watch can easily measure skin temperature, but their readings often deviate significantly from true CBT [4]. A wearable solution is clearly needed that enables unobtrusive, continuous, and high-fidelity CBT tracking in daily life.

Moreover, existing applications have only begun to unlock the full potential of continuous CBT monitoring. CBT fluctuates with health condition, physical activity, meals, sleep cycles, and emotional stress. Yet most systems merely display raw temperature curves without decomposing these complex drivers [18]. To enable truly personalized well-being, each behavior's contribution to observed CBT changes needed to be decomposed and quantified.

This paper presents a skin-attachable CBT sensor and a companion smart device app that together address these challenges. The sensor incorporates a thermodynamically optimized structure within a compact patch to minimize environmental heat flux loss and enable CBT to be accurately estimated. The app correlates continuous CBT traces with user activities and provides contextualized feedback for health optimization. Our main contributions are as follows. First, we have developed a compact and universal skin-attachable device that integrates skin temperature and heat flux measurements to estimate CBT with high precision. Second, we have also developed a contextual smart device app for real-time CBT visualization and personalized health recommendations in daily life. Finally, we have realized activity inference from sensor dynamics by leveraging high-frequency thermal *noise* to decompose CBT changes into behavioral drivers without auxiliary sensors. The live demonstration presenting these contributions allows participants to experience how a single easy-to-use CBT sensor enables diverse applications for personal health and daily living.

2 Related Work

Various noninvasive methods have been explored to measure CBT in everyday settings. Clinical-grade approaches—rectal probes, tympanic thermometers, and oral thermometers—provide high accuracy [13], but their invasiveness and discomfort make them impractical for long-term use in daily life. Ingestible capsule sensors [19] improve practicality by allowing unrestricted physical activity, yet they suffer from unpredictable gastrointestinal transit times, data retrieval challenges, and single-use disposal. Skin-attachable sensors [9] offer a promising noninvasive alternative for extended monitoring, but their accuracy often degrades due to environmental heat flux losses and ambient variability. Focusing on comfort and design, Thermal Earring [18] adopts an earlobe-worn form factor to collect peripheral temperature continuously, but it only measures skin temperature trends rather than CBT. Moreover, none of these approaches decomposes the behavioral drivers underlying temperature fluctuations.

While these sensor designs and applications have achieved important milestones, none combine high accuracy, long-term unobtrusiveness, and behavioral analysis in a single system. Our work builds on these foundations by combining a skin-attachable CBT sensor with a thermodynamically optimized structure and a context-aware smartphone app. Our system estimates CBT with $\pm 0.1^\circ\text{C}$ accuracy in laboratory tests and decomposes temperature fluctuations into actionable insights tied to daily routines. By unifying precision, wearability, and contextual interpretation, we advance CBT sensing toward personalized healthcare optimization.



Figure 2: Size comparison of the skin-attachable CBT sensor (about 30 mm diameter) and a 1-euro coin: demonstrating the device's compact form factor for comfortable everyday skin attachment.

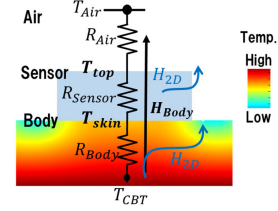


Figure 3: Thermal circuit model for CBT estimation in the skin-attachable sensor [16]: The color map shows temperature distribution in the body region, with T_{air} , T_{skin} , and T_{top} denoting ambient, skin-surface, and sensor-top temperatures, and T_{CBT} denoting CBT. The optimized structure reduces the lateral heat flux H_{2D} .

3 CBT Sensor

Figures 2 and 3 illustrate the appearance and operating principle of our CBT sensor, respectively. The device is a compact skin patch measuring 30 mm in diameter and 5 mm in thickness. Its sensing principle is based on a one-dimensional heat-conduction model [15, 16] derived from Pennes's bioheat equation [12]. A truncated cone structure made of a material with high thermal conductivity reduces lateral heat flux and enhances CBT estimation accuracy [15].

CBT T_{CBT} is modeled as

$$T_{\text{CBT}} = T_{\text{skin}} + R_{\text{body}} H_{\text{body}}, \quad (1)$$

where T_{skin} denotes the skin-sensor interface temperature, H_{body} the conductive heat flux from the body core, and R_{body} a constant coefficient related to the thermal properties of the body and the sensor. To measure T_{skin} and H_{body} passively, two thermistors are embedded: one at the skin interface (reading T_{skin}) and one atop a heat-resistant substrate (reading T_{top}). Since the heat flux is then proportional to the temperature difference, $H_{\text{body}} \propto T_{\text{skin}} - T_{\text{top}}$, substitution into (1) yields real-time CBT estimation as

$$T_{\text{CBT}} = T_{\text{skin}} + R'_{\text{body}} (T_{\text{skin}} - T_{\text{top}}), \quad (2)$$

where R'_{body} is determined during an initial calibration using a reference measurement (e.g., tympanic).

An important advantage of our sensor is its placement flexibility, enabling easy attachment at any skin location for universal use. Since R'_{body} , derived from R_{body} , varies with both the individual and the attachment site, initial calibration is required for precisely estimating CBT. Alternatively, if only the overall CBT change pattern is of interest, measurements can proceed without individual calibration by using default parameters for each sensor location [16].

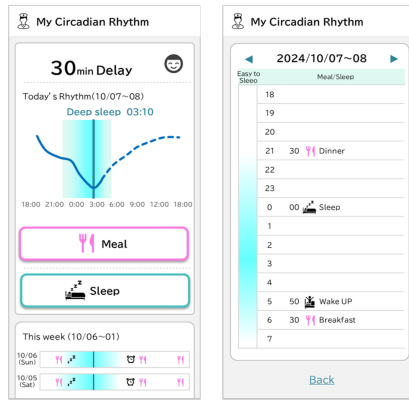


Figure 4: Main screens of the companion app visualizing circadian rhythm: (Left) Estimated nadir view alongside their typical or ideal rhythm. (Right) and a daily timeline view showing logged activities with sleep depth.

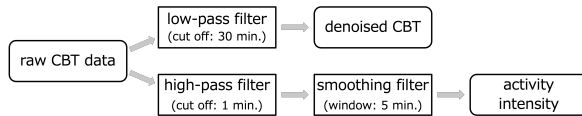


Figure 5: Data-processing workflow for CBT and activity extraction: Raw CBT data is split into two streams: a low-pass filter for denoising, and a high-pass filter followed by a smoothing filter to derive an activity intensity index.

4 CBT Smart Application

4.1 Smartphone App Interface

To facilitate everyday use of continuous CBT data, our companion app connects to the CBT sensor to visualize users' daily CBT traces and derived metrics. The app aims to support lifestyle improvement by visualizing CBT rhythms and fluctuations in real time and linking them to daily behaviors. Figure 4 illustrates the app's main functional screens. The top section of the left screen displays the user's circadian rhythm and nadir (the lowest core temperature during sleep). Users can compare their current rhythm against their typical or ideal pattern. Additionally, daily activities closely related to CBT (such as sleep, meals, and exercise) are logged and displayed together on the right screen. By presenting all data together, the app enables users to review and optimize CBT-based lifestyle patterns encompassing not only sleep but also other daily activities.

4.2 Decomposition of Daily-Life CBT Changes

CBT fluctuates due to various physiological and environmental factors, and its pattern differs among individuals. To leverage these fluctuations for lifestyle optimization, it is essential to understand not only individual CBT traces but also which behaviors cause these changes. This paper extracts activity intensity by leveraging high-frequency noise in 1 Hz CBT data without additional hardware. The body's high thermal capacitance produces slow CBT variations on timescales of tens of minutes to hours. In contrast, rapid fluctuations at 1-sec to 1-min intervals reflect measurement and environmental

noise rather than true CBT changes. To extract both the long-term trend and short-term variability, we apply two complementary filters to a 1 Hz sampled CBT signal.

Figure 5 depicts the data-processing workflow. First, a low-pass filter with a 30-min cutoff is applied to raw CBT data to extract a denoised CBT trace. Simultaneously, a high-pass filter with a 1-min cutoff is applied to the raw data to extract only the measurement noise, which is then smoothed over 5 min to reflect human behavioral time scales. This processed signal serves as an activity intensity index that indirectly reflects physical movement without additional hardware such as accelerometers.

5 Trial Experiment in the Wild

A trial experiment in the wild was conducted with four consenting adult participants using the developed CBT sensors. The experiment aimed to validate two factors: 1) whether the sensors can unobtrusively capture CBT traces across daily activities and attachment sites and 2) whether the activity intensity index enables the behavioral drivers behind CBT change to be identified.

5.1 Trial Settings

Participants wore CBT sensors at three skin sites: left forehead edge (Forehead), above the left collarbone (Neck), and under the left armpit (Armpit) throughout their normal daily routines. These sites were selected to reflect different real-world usage conditions including movement stability, clothing interference, and skin contact variability. Each measurement session began after the end of the post-bathing rest period and ended the next day's pre-bathing period. At the start of each session, a tympanic temperature reading was taken for calibration. Reference data including step count, heart rate, and sleep duration were collected using a Fitbit Versa 4. In addition, participants self-logged the time of meals, exercise, and sleep as annotations. If participants experienced discomfort or skin irritation, they could pause measurements at affected sites or times. This occasionally led to differences in data completeness across sensor sites for the same events.

5.2 Experimental Results

Figure 6 presents raw and filtered data including annotations from a single measurement session of one participant. The denoised CBT traces exhibit similar patterns across sites, although exercise-induced peak amplitudes and baseline fluctuations varied, possibly due to differences in clothing insulation. Furthermore, all sensors reliably detected CBT increases associated with meals and exercise even during a short period of activity. Note that a sudden drop to about 35°C on Neck sensor showed partial detachment during that period.

The activity intensity index exhibited larger peaks during exercise than during meal times in Figure 6. For deeper analysis, Fitbit step counts showed substantial movement during annotated exercise sessions and revealed additional unlogged activity, such as household tasks. Figure 7 shows a scatter plot of activity intensity and step counts for the forehead sensor in the same day. The Pearson correlation coefficient ($r = 0.662$) indicates a strong positive relationship between the index and the intensity of physical activity.

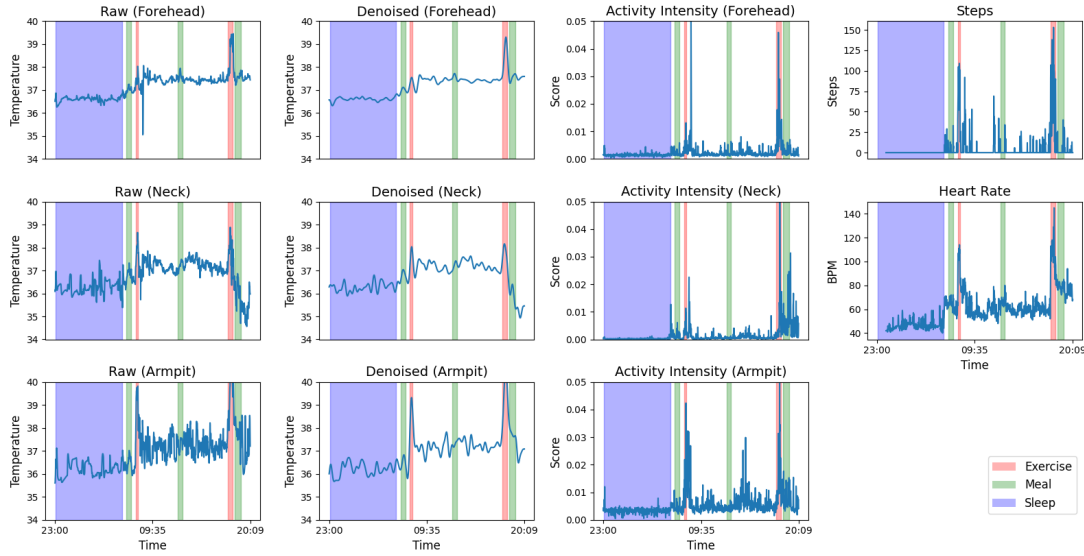


Figure 6: One-day CBT recordings from forehead, neck, and armpit: raw CBT (col 1), denoised CBT (col 2), and activity intensity (col 3) with activity annotations. Fitbit step counts and heart rate (col 4) provide reference. These results demonstrate that CBT rises coincide with activities, but the activity intensity index differs between exercise and meals.

Table 1: Comparison of metrics between Meal and Exercise across body positions

Metric	Forehead		Neck		Armpit	
	Meal	Exercise	Meal	Exercise	Meal	Exercise
# Events	26	14	28	14	29	16
ΔCBT [$^{\circ}\text{C}$]	0.34	0.86	0.45	0.67	0.25	2.49
Activity Intensity	0.0021	0.0095	0.0049	0.0250	0.0046	0.0160
Steps	1.50	57.8	3.19	57.5	1.88	57.5

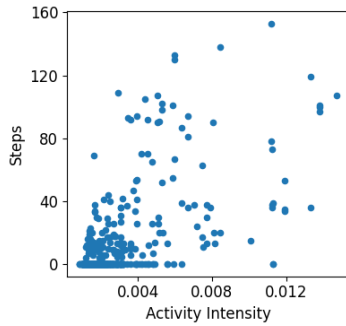


Figure 7: Correlation between activity intensity and Fitbit step count in a day: the overall data show a positive correlation ($r = 0.662$) regardless of activity annotation.

For quantitative evaluation in whole data, Table 1 compares CBT change, activity intensity, and step count between meal and exercise across the three sites. These results demonstrate that our system not only captures daily behaviors affecting CBT but also distinguishes physical activity-driven changes from non-activity-driven ones by incorporating the activity intensity index.

6 Demonstration

In the live demo, participants can personally handle and attach the sensors and immediately visualize how brief health behaviors affect their CBT. In addition, they can see long-term CBT visualization across different body sites using multiple sensors, accompanied by an onsite technical overview. Through this demonstration, we show that a single, easy-to-use CBT sensor system can support diverse applications for enhancing personal daily health.

7 Conclusion

This paper presents a smart system that can measure core body temperature (CBT) over a long term without any restrictions on the position of the users or their behavior and visualize their life rhythms. Through measurements in the wild in daily life without any restrictions, this system not only visualized CBT but also showed the possibility of identifying whether the change factor of CBT is physical activity or not by CBT oversampling process. In the demonstration, the participants can touch the sensor system by themselves, which gives them the expectation that the system can enhance their daily life in various situations. In future work, we will further analyze the factors of CBT changes such as during sleep and develop an effective intervention application.

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